

Adil Saeed

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AI/ML Engineer specializing in Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), NLP, and Computer Vision. Experienced in building production-ready AI applications using Python, PyTorch, TensorFlow, Hugging Face, FAISS, OpenAI APIs, and Gradio. Passionate about scalable AI systems for education, healthcare, and enterprise solutions.

EDUCATION

Bachelor of Science in Software Engineering CGPA: 3.7/4.0
Institute of Management Sciences (IMSciences), Peshawar

ML-to-LLM Bootcamp Summer 2025
Ghulam Ishaq Khan Institute (GIKI), Topi, Pakistan

LEADERSHIP & EXPERIENCE

AI/ML Lead 2024 – Present
Microsoft Learn Student Ambassadors

- Led technical workshops on ML, DL, and NLP, training 50+ students in AI project development.
- Mentored peers in building scalable ML applications, highlighting ethical AI practices.
- Organized boot camps to promote AI adoption in academia and industry.

Ambassador – Institute of Emerging Careers

Digital Architect Conference

2025 – Present

- Represented IMSciences at high-level conferences, engaging with international speakers and industry experts at 20+ sessions.
- Moderated diverse sessions, gaining exposure to global perspectives in AI, digital transformation, and emerging tech.
- Supported event coordination and participant engagement to enhance learning outcomes.

Project Intern

Ghulam Ishaq Khan Institute (GIKI), Topi, Pakistan

Summer 2025

- Developed a Retrieval-Augmented Generation (RAG) chatbot for GIKI prospectus queries.
- Collaborated on applied ML projects, including NLP and vector search.
- Assisted in technical workshops to enhance participants' AI skills.

KEY PROJECTS

GIKI Prospectus Q&A Chatbot | [GitHub](#)

Python • FastAPI • FAISS • Hugging Face • OpenAI API • Gradio

- Developed a multilingual (English/Urdu) Retrieval-Augmented Generation (RAG) chatbot for answering GIKI prospectus queries.
- Implemented document parsing, embeddings, FAISS vector search, and context-aware retrieval using OpenAI APIs.

- Built an interactive Gradio interface with chat export and deployed the application for internship demonstrations.

Sentiment Analysis System | [GitHub](#)

Python • Scikit-learn • TF-IDF • Logistic Regression

- Built a sentiment classification system using TF-IDF and Logistic Regression, achieving 85% precision.
- Performed text preprocessing, feature engineering, and model evaluation for improved classification performance.
- Designed a reusable prediction pipeline for real-world text inputs.

Face Recognition System | [GitHub](#)

Python • OpenCV • CNN

- Developed a real-time face recognition system using CNNs and OpenCV.
- Improved recognition accuracy through transfer learning and dataset augmentation.
- Optimized image preprocessing for faster and more reliable inference.

CardioDiagnosis – Heart Disease Prediction | [GitHub](#)

Python • Scikit-learn • Random Forest • SVM

- Developed machine learning models for predicting heart disease risk with 90% recall.
- Evaluated model performance using ROC curves, cross-validation, and classification metrics.
- Compared multiple algorithms to identify the best-performing predictive model.

Crop Yield Prediction System | [GitHub](#)

Python • Scikit-learn • Regression Models

- Developed an end-to-end machine learning pipeline for crop yield prediction using historical agricultural data.
- Performed feature engineering, hyperparameter tuning, and model evaluation.
- Compared regression models to improve prediction accuracy.

Login Risk Detection System | [GitHub](#)

Python • Scikit-learn • Random Forest

- Built an anomaly detection model for identifying high-risk login attempts.
- Applied feature engineering to improve fraud detection performance.
- Evaluated model accuracy using classification metrics and confusion matrices.

TECHNICAL SKILLS

- **Programming:** Python, C++, Java
- **Machine Learning & AI:** Scikit-learn, TensorFlow, PyTorch, OpenCV
- **Generative AI & NLP:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Hugging Face Transformers, FAISS, Prompt Engineering, Context Management.
- **Backend & APIs:** FastAPI, REST APIs, Gradio, Streamlit
- **Web Technologies:** HTML, CSS, JavaScript, Bootstrap
- **Deployment:** Vercel, Railway, Hugging Face Spaces
- **Databases:** MySQL, SQL
- **Tools & Platforms:** Git, GitHub, Jupyter Notebook, Google Colab, VS Code, Linux (Basic CLI)

Research Interests

- Large Language Models (LLMs)
- Retrieval-Augmented Generation (RAG)
- Multilingual NLP
- AI Agents
- Responsible AI

LANGUAGES

English – Fluent

- Urdu – Native
- Pashto-Fluent

PUBLICATIONS & WRITING

Medium Technical Writer (2024 – Present)

Published articles on: [Medium](#)

- *Domain Expertise: Coding in Modern AI*
- *Why Aspiring Generative AI Engineers Should Learn with Google Gemini*
- *From Research to Production with PyTorch*
- *When Your RAG Retriever Lies to You: Understanding Corrective RAG (CRAG)*
- *Understanding Hypothetical Retrieval: The Next Step Toward Smarter AI Systems*

Writing frequency: One article every 1–2 months.

